

Konstantinos Zoropoulos

43, Korai str.17123 N. Smyrni

+30 6988633078

kostaszoropoulos@gmail.com

Computer Engineering undergraduate specializing in low-level programming, RISC-V architecture, and embedded systems, seeking an industry internship in hardware-software integration.

PROJECTS

Fall Detection System Prototype — *Personal*

-Architecting a hardware prototype utilizing ESP32 microcontrollers and mmWave radar sensors for reliable, non-intrusive fall detection.

-Formulating the commercial development strategy and system integration pipeline.

RISC-V Extension for Visual Studio — *Personal*

-Developing an open-source extension for the Microsoft Visual Studio ecosystem.

-Implemented support for RISC-V assembly code compilation and execution via the RARS application environment.

Modular Arithmetic Calculator — *Personal*

-Designed a fully functional, battery-powered calculator built on the Arduino microcontroller platform.

-Executed custom 3D-printing for the chassis, manual hardware wiring, and the integration of a modular user interface.

Custom SQL Lexer and Parser — *Academic*

-Engineered a custom lexical analyzer and parser utilizing the combination of Flex, Bison, and C.

-Designed strict grammatical rules for the accurate interpretation and robust error handling of SQL instructions.

Data Analysis Application — *Academic*

-Developed a Python-based application utilizing REST APIs and web scraping techniques to collect course data.

-Implemented algorithms to analyze, sort, and visualize the aggregated information within a custom Graphical User Interface.

EDUCATION

Computer Engineering & Informatics Department (CEID), University of Patras — *Undergraduate*

2024 - Present

-**Current GPA: 7.2 / 10.0**

-**Relevant Coursework:** Data Structures, Signal and System Theory, Numerical Analysis, Computer Architecture.

EXPERIENCE

Kumon Study Center, Glyfada — *Assistant Educator in Mathematics & English*

October 2023 - September 2024 & July 2025 - September 2025

SKILLS

Programming Languages: C, C++, Python, Java, HTML, JavaScript.

Hardware & Architecture: RISC-V ISA, Assembly, Datapath Design, ESP32, mmWave sensors.

Systems & Tools: Linux System Administration (Fedora, Ubuntu, Pop!_OS), Advanced Terminal/CLI Usage, GNU Nano, Git.

SEMINARS

CreaTech Project - European Institute of Technology HEI

-Knowledge Lab Training Program focused on creative industries business knowledge and business plan creation.

LANGUAGES

English: C2 Level (Certificate of Proficiency in English, University of Michigan).

Greek: Native.